

V. DISCUSSION

A. Applicability to Different Multi-Beam LEO Constellations

The proposed EABR framework is designed for general multi-beam LEO satellite systems and is not limited to a specific constellation. Although the main service-performance evaluation in this work uses a OneWeb-like constellation, the core idea of EABR depends on predictable satellite motion, beam footprints, beam-level EPFD contributions, and coverage overlap among neighboring LEO satellites. These properties are common in many multi-beam LEO systems.

Therefore, as long as the satellite geometry and beam coverage can be estimated in advance, the time-slot beam role record and candidate beam relation graphs can be constructed for different constellation configurations. The proposed framework can then identify the critical satellites and helper satellites according to the current system state. In this sense, EABR is applicable to different multi-beam LEO constellations. In addition, helper satellite availability is examined under OneWeb-like, Starlink-like, and Lightspeed-like geometries to determine whether helper satellite candidates are generally available under different constellation configurations.

B. Helper Availability and Resource-Limited Recovery

Helper-based service recovery first requires neighboring satellites whose beams overlap with the closed beams of critical satellites. The availability of these helper satellite candidates depends on the constellation geometry, including the satellite distribution, orbital spacing, and beam footprint overlap. Section VI further examines helper satellite availability under different constellation geometries, and the results indicate that helper satellite candidates are generally available under the evaluated constellation geometries.

Even when helper satellite candidates are available, the recovery performance of EABR is still limited by the available helper beam capacity and helper satellite transmit power. Therefore, insufficient recovery resources should be regarded as a system-level resource limitation rather than a failure of the reassociation framework. As long as feasible helper coverage and available power exist, EABR can use these resources for service recovery while maintaining the EPFD and power constraints.

C. Priority-Aware Recovery and Starvation Risk

The HBR recovery score incorporates user priority weights to ensure that users with higher service priorities receive greater importance during service recovery. Under high-load conditions, the available helper recovery capacity may be insufficient to serve all closed-beam users. In this case, among the candidate closed-beam user subsets evaluated for each closed-recovery edge, subsets containing more high-priority users are more likely to obtain higher recovery scores and be selected first. Consequently, some low-priority users may remain unrecovered for several consecutive time slots, resulting in potential user starvation. This behavior reflects a tradeoff between priority-aware service recovery and user fairness. Future work may introduce a starvation-prevention mechanism to mitigate or avoid low-priority user starvation.

REFERENCES

- [1] M. Giordani and M. Zorzi, "Non-terrestrial networks in the 6g era: Challenges and opportunities," *IEEE Network*, vol. 35, no. 2, pp. 244–251, 2021.
- [2] X. Lin, S. Cioni, G. Charbit, R. Chuberre, S. Hellsten, and J.-F. Boutillon, "On the path to 6g: Embracing the next wave of low earth orbit satellite access," *IEEE Communications Magazine*, vol. 59, no. 5, pp. 36–42, 2021.
- [3] X. Zhu and C. Jiang, "Integrated satellite-terrestrial networks toward 6g: Architectures, applications, and challenges," *IEEE Internet of Things Journal*, vol. 9, no. 1, pp. 437–461, 2022.
- [4] O. B. Al-Hraishawi, H. Chaaban, S. Kisseleff, E. Lagunas, and S. Chatzinotas, "A survey on non-geostationary satellite systems: The communication perspective," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 1, pp. 101–132, 2023.
- [5] ITU-R, "Radio regulations," International Telecommunication Union, Geneva, Switzerland, 2020, see Article 22 for NGSO–GSO interference limits and EPFD framework.
- [6] —, "Functional description to be used in developing software tools for determining conformity of non-geostationary-satellite orbit fixed-satellite service systems or networks with limits contained in article 22 of the radio regulations," International Telecommunication Union Radiocommunication Sector (ITU-R), Geneva, Switzerland, Recommendation S.1503-3, 2018.
- [7] International Telecommunication Union Radiocommunication Sector (ITU-R), "Satellite antenna radiation patterns for non-geostationary orbit satellite antennas operating in the fixed-satellite service below 30 ghz," ITU-R, Geneva, Switzerland, Recommendation S.1528, 2001.
- [8] ITU-R, "Reference fss earth-station radiation patterns for use in interference assessment involving non-gso satellites in frequency bands between 10.7 ghz and 30 ghz," ITU-R, Geneva, Switzerland, Recommendation S.1428-1, 2001.
- [9] —, "Calculation of free-space attenuation," International Telecommunication Union Radiocommunication Sector, Geneva, Switzerland, Recommendation P.525.
- [10] —, "Attenuation by atmospheric gases and related effects," International Telecommunication Union Radiocommunication Sector, Geneva, Switzerland, Recommendation P.676.
- [11] C. Braun, A. M. Vojcic, L. Simic, and P. Mähönen, "Should we worry about interference in emerging dense ngso satellite constellations?" in *Proceedings of the IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN)*, 2019, iEEE Xplore document 8935875.
- [12] A. Hills, J. M. Peha, and J. Munk, "Feasibility of using beam steering to mitigate ku-band leo-to-geo interference," *IEEE Access*, vol. 10, pp. 74 023–74 032, 2022.
- [13] M. Jalali, F. Ortiz, E. Lagunas, S. Kisseleff, L. Emiliani, and S. Chatzinotas, "Joint power and tilt control in satellite constellation for ngso–gso interference mitigation," *IEEE Open Journal of Vehicular Technology*, 2023, early access / accepted version available via IEEE Xplore.
- [14] T. Li, J. Jin, W. Li, Z. Ren, and L. Kuang, "Research on interference avoidance effect of onweb satellite constellation's progressive pitch strategy," *International Journal of Satellite Communications and Networking*, pp. 1–15, 2021.
- [15] G. Cocco, M. Angelone, and A. I. Pérez-Neira, "Co-channel interference cancellation at the user terminal in multibeam satellite systems," in *Proceedings of the IEEE International Conference on Communications (ICC)*, 2014, pp. 3992–3997.
- [16] C. E. Shannon, "A mathematical theory of communication," *Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 1948.
- [17] J. Lei and M. A. Vázquez-Castro, "Multibeam satellite frequency/time duality study and capacity optimization," *Journal of Communications and Networks*, vol. 13, no. 5, pp. 472–480, 2011.
- [18] WorldVu Satellites Limited, "Onweb non-geostationary satellite system," FCC IBFS File No. SAT-LOI-20160428-00041, Technical Attachment, Washington, DC, USA, 2016, oneWeb NGSO system description.
- [19] Analytical Graphics, Inc., "Systems tool kit (stk)," <https://www.agi.com/products/stk>, 2023, online; accessed 21-Jul-2025.
- [20] The MathWorks, Inc., "Matlab r2023b," <https://www.mathworks.com/products/matlab.html>, 2023, online; accessed 21-Jul-2025.